

FINITE COLLATZ HISTORIES THROUGH SUPPORT AND HURWITZ ZERO CLUSTERS

KOKUNO YUMETO

ABSTRACT. Finite Collatz itineraries admit a polynomial affine lift that retains the positions of all odd steps. We give its inverse, its residue realization, and explicit fibres of several forgetful maps. We then apply the Split Zero programme's compact Hurwitz reconstruction to exact odd-return residue coordinates. On a known packet of K words, the full local zero-cluster jet through order $K - 1$ recovers every probability weight and every joint arithmetic predicate on the words. The inverse is explicit, works at a nontrivial Riemann zeta zero of any multiplicity, and has a sharp worst-case jet order. A three-word calculation recovers integral-cycle candidate mass from the first two moments. These are finite information-preserving constructions; they do not supply an orbit-descent estimate.

1. THE QUESTION AND ITS SOURCE

The Split Zero construction distinguishes an absent component from a present component whose amplitude is zero. Its Collatz F-series note [1] already represents affine branches on support-homogeneous coordinates. Its later Hurwitz reconstruction [2] recovers a compact probability measure from a complete local zero cluster. The question here is what these constructions retain when their inputs are actual finite Collatz histories.

There are two steps. First, history must be attached to faithful coordinates: a label beside a collapsed scalar cannot restore an arbitrary lost word. Second, the spectral family must be specified. We use a Hurwitz deformation with an explicit inverse, rather than the eigenvalues of the two-by-two branch product. The latter have a computable loss of information. Classical parity coding supplies the dynamical coordinates; neither the affine composition identity nor finite parity coding is claimed as new. The contribution of this note is their explicit finite transport through the cited reconstruction, with synchronized arithmetic tests retained.

2. A FAITHFUL POLYNOMIAL LIFT OF A FINITE HISTORY

Let q be an indeterminate and $R = \mathbb{Z}[1/2][q]$. For $d \in \{0, 1\}$ put

$$T_d^{(q)}(x) = \frac{q^d x + d}{2}.$$

At $q = 3$, selecting d to be the parity of the current integer gives the shortened Collatz map. A word $w = (d_0, \dots, d_{N-1})$ is read in chronological order; T_{d_0} acts first. Write

$$m = \sum_{j=0}^{N-1} d_j, \quad s_j = \sum_{i < j} d_i, \quad P_w(q) = \sum_{j=0}^{N-1} d_j 2^j q^{m-1-s_j},$$

where a summand with $d_j = 0$ is omitted, so no negative exponent occurs.

Proposition 2.1 (Lift, inverse, and realization). *For every word w ,*

$$(1) \quad T_{d_{N-1}}^{(q)} \circ \dots \circ T_{d_0}^{(q)}(x) = \frac{q^m x + P_w(q)}{2^N}.$$

Date: 13 September 2026. Research-program note.

The map $w \mapsto (N, P_w)$ is injective. Its image consists of $(N, 0)$ and pairs

$$\left(N, \sum_{k=0}^{m-1} 2^{j_k} q^{m-1-k}\right), \quad 0 \leq j_0 < \cdots < j_{m-1} < N.$$

The inverse places ones precisely at the j_k and zeros elsewhere. A positive integer n realizes w as its first N shortened-map parities exactly when

$$(2) \quad 3^m n + P_w(3) \equiv 0 \pmod{2^N}.$$

Thus each word occurs on one residue class modulo 2^N .

Proof. Appending d replaces (m, P) by $(m + d, q^d P + d2^N)$, which proves (1) and the coefficient formula by induction. If the one positions are $j_0 < \cdots < j_{m-1}$, the coefficient of q^{m-1-k} is exactly 2^{j_k} . This proves the image and inverse, including the all-zero word. The length is essential: terminal zero bits contribute no new coefficient to P .

For realization use induction on N . An actual orbit satisfies (2) by (1). Conversely suppose the congruence holds. If $d_0 = 0$, then $P_w(3)$ is even, so n is even; write $n = 2y$ and $P_w(3) = 2P_{w'}(3)$ for the remaining word w' . Division by two yields its congruence at y . If $d_0 = 1$, then $P_w(3) = 3^{m-1} + 2P_{w'}(3)$ is odd, so n is odd; put $y = (3n + 1)/2$. The same division gives $3^{m-1}y + P_{w'}(3) \equiv 0 \pmod{2^{N-1}}$. Induction proves every requested parity. The coefficient 3^m is invertible modulo 2^N , giving the unique residue. Positive representatives exist and remain positive along these actual iterates. $\square$

The source F-series uses the opposite matrix order, with d_0 acting last on zero:

$$X_w(q) = T_{d_0}^{(q)} \circ \cdots \circ T_{d_{N-1}}^{(q)}(0) = \sum_{j=0}^{N-1} \frac{d_j q^{s_j}}{2^{j+1}}.$$

If w^{rev} reverses the word, then

$$X_{w^{\text{rev}}}(q) = 2^{-N} P_w(q).$$

Indeed, reversing a one at j puts it at $N - 1 - j$, with $m - 1 - s_j$ ones before it. This is the exact map between the two composition conventions, not an identification of input binary digits with orbit parities.

2.1. Support, specialization, and what each forgets. For the source semiring $G(R) = R \sqcup \{\tau\}$, τ is the additive identity and multiplicative absorber, while $e = 0_R$ is a supported zero. For $r, s \in R$, their addition and multiplication are those of R . The reflection $p : G(R) \rightarrow R$ fixes R and sends τ to 0. Its zero fibre is exactly $\{\tau, e\}$; every other fibre is a singleton. The Boolean support character has

$$\chi(\tau) = 0, \quad \chi(r) = 1 \quad (r \in R),$$

including $\chi(e) = 1$. Hence this character detects presence, not arithmetic nonvanishing or parity. For a nonzero coefficient ring, the source branch matrices are

$$M_0 = \begin{pmatrix} 1/2 & e \\ \tau & 1 \end{pmatrix}, \quad M_1 = \begin{pmatrix} q/2 & 1/2 \\ \tau & 1 \end{pmatrix}.$$

Both have the same Boolean support matrix. Their full polynomial products retain more than this support projection.

To make a history mask explicit, set $A_N = R^{\{0, \dots, N-1\}}$ and $\epsilon_w = (d_0, \dots, d_{N-1}) \in A_N$. It is a central idempotent. Its j th coordinate is the parity bit itself, so (N, ϵ_w) recovers the word. This mask is additional retained data; applying χ to the two branch matrices does not produce it.

The evaluation homomorphism $R \rightarrow \mathbb{Z}[1/2]$, $q \mapsto 3$, has kernel $(q - 3)$ by division by the monic polynomial $q - 3$. For the source F-series, take the five-bit words with one positions $\{0, 4\}$ and $\{2, 3, 4\}$ respectively. Then

$$X_u(q) = \frac{1}{2} + \frac{q}{32}, \quad X_v(q) = \frac{1}{8} + \frac{q}{16} + \frac{q^2}{32},$$

and

$$X_v(q) - X_u(q) = \frac{(q-3)(q+4)}{32}, \quad X_u(3) = X_v(3) = \frac{19}{32}.$$

These are distinct histories in one specialization fibre. Under $G(R) \rightarrow G(R/(q-3))$ their nonzero difference becomes the supported zero e , not τ . Keeping that support label records that a contribution was present and killed; recovering the contribution itself requires its preimage or the retained polynomial coefficients.

The reflected forward product in (1) has matrix

$$\begin{pmatrix} q^m/2^N & P_w(q)/2^N \\ 0 & 1 \end{pmatrix}.$$

Its eigenvalue multiset is $\{q^m/2^N, 1\}$, independent of the order of the bits. At $N = 2, m = 1$, $w = (0, 1)$ has translation $1/2$ and $w = (1, 0)$ has translation $1/4$, with the same eigenvalues. This computes the loss for this spectral map; it does not preclude a different, faithful spectral construction.

3. EXACT ODD-RETURN COORDINATES

Fix integers $m \geq 1$ and $A \geq m$, and let

$$W_{m,A} = \{(a_1, \dots, a_m) \in \mathbb{N}_{>0}^m : \sum a_i = A\}, \quad K = |W_{m,A}| = \binom{A-1}{m-1}.$$

These are labelled words, not rotation classes. Put $S_j = a_1 + \dots + a_j$, $S_0 = 0$, and

$$B_w = \sum_{j=0}^{m-1} 3^{m-1-j} 2^{S_j}, \quad D = 2^A - 3^m.$$

The odd-return branch composition is $(3^m x + B_w)/2^A$. Let r_w be the unique representative $1 \leq r_w < 2^{A+1}$ satisfying

$$(3) \quad 3^m r_w + B_w \equiv 2^A \pmod{2^{A+1}}.$$

Lemma 3.1. *The positive odd integers with first m exact odd-return exponents w are precisely $r_w + 2^{A+1}\mathbb{Z}$, intersected with the positive integers. The residues r_w are pairwise distinct on $W_{m,A}$.*

Proof. For one letter, (3) is exactly $\nu_2(3x+1) = a_1$. For more letters write $B_w = 3^{m-1} + 2^{a_1} B_{w'}$. The congruence first implies $2^{a_1} \mid 3x+1$. Put $y = (3x+1)/2^{a_1}$ and divide by 2^{a_1} . The resulting congruence is

$$3^{m-1}y + B_{w'} \equiv 2^{A-a_1} \pmod{2^{A-a_1+1}}.$$

Since $A - a_1 \geq 1$ and $B_{w'}$ is odd, it forces y odd. The valuation was therefore exact, and induction applies to the suffix. The converse follows by reversing this calculation. Multiplication by 3^m is invertible modulo 2^{A+1} , proving uniqueness and oddness of r_w . If two words shared a residue, its positive representative would have two different exact exponent lists, impossible for a function. $\square$

The rational periodic candidate is $x_w = B_w/D$. The denominator is nonzero, odd, and coprime to three. Rotation σ satisfies

$$3B_w + D = 2^{a_1} B_{\sigma w}.$$

Because all B are odd, this gives exactly the exponent a_1 in the rationals with odd denominator. The candidates are positive when $D > 0$ and integral precisely when $D \mid B_w$. In the latter case the identity and coprimality propagate integrality around the word. These observations justify the predicate used below without mistaking all rational candidates for integer cycles.

4. A ZERO-CLUSTER INVERSE FOR PACKET DISTRIBUTIONS

Let $\Delta(W)$ be the probability simplex on the known word set $W = W_{m,A}$. For $\lambda \in \Delta(W)$ define, in the original integer residue coordinate,

$$(4) \quad \mu_\lambda = \sum_{w \in W} \lambda_w \delta_{r_w}, \quad M_j = \sum_w \lambda_w r_w^j, \quad Z_\lambda(s, t) = \sum_w \lambda_w \zeta(s, 1 + tr_w).$$

Here $M_0 = 1$. Take $L = 2^{A+1} - 1$ and $|t|L < 1$. The parameter t is a complex deformation parameter, not Collatz time. Hurwitz Taylor expansion [4] gives

$$(5) \quad Z_\lambda(s, t) = \sum_{j \geq 0} M_j b_j(s) t^j, \quad b_j(s) = \frac{(-1)^j}{j!} (s)_j \zeta(s + j).$$

On a compact subset of $0 < \Re s < 1$, the factor $(s)_j/j!$ is bounded by a polynomial in j : multiply $1 + (s-1)/k$ for $1 \leq k \leq j$ and use the harmonic-sum bound on its absolute value. For $j \geq 1$ the zeta factor is uniformly bounded by its absolutely convergent Dirichlet series. Since $|M_j| \leq L^j$, the series is normally convergent locally for $|t|L < 1$. This establishes joint holomorphicity in the neighbourhoods used below. At any nontrivial zero ρ , every $b_j(\rho)$, $j \geq 1$, is nonzero: the rising factorial has no zero factor, and $\Re(\rho + j) > 1$, where the absolutely convergent Euler product makes zeta nonvanishing.

Lemma 4.1 (Cluster-to-moment inverse, after the Split Zero reconstruction). *Fix a nontrivial zero ρ of exact multiplicity r . Let $Q_\lambda(s, t)$ be the monic degree- r polynomial in $s - \rho$ whose roots are the entire nearby zero cluster of $Z_\lambda(\cdot, t)$, with multiplicities. Then its t -jet through order J and $(M_1, \dots, M_J)$ determine one another.*

Proof. Choose a disc wholly inside the critical strip that contains only ρ among the zeros at $t = 0$, with no boundary zero. Uniform convergence and Rouché's theorem give precisely r interior zeros for small t . The contour integrals of $(s - \rho)^k \partial_s Z_\lambda / Z_\lambda$, divided by $2\pi i$, give the power sums of their displacements, for $1 \leq k \leq r$. Newton's identities construct holomorphic elementary symmetric functions and hence Q_λ , without choosing individual branches. Write

$$Q_\lambda = (s - \rho)^r + \sum_{j \geq 1} Q_j(s) t^j, \quad Z_\lambda = U_\lambda Q_\lambda, \quad U_\lambda = \sum_{j \geq 0} U_j(s) t^j.$$

The quotient is jointly holomorphic: on an enclosing contour it is holomorphic in t , and its Cauchy integral gives its extension in s . For each fixed t the two divisors coincide, so all apparent singularities are removable. It is a unit, and $U_0 = \zeta(s)/(s - \rho)^r$ is known. Coefficient comparison yields the successive inverse formulas

$$(6) \quad M_j = \frac{\sum_{k=1}^j U_{j-k}(\rho) Q_k(\rho)}{b_j(\rho)},$$

$$(7) \quad U_j(s) = \frac{M_j b_j(s) - \sum_{k=1}^j U_{j-k}(s) Q_k(s)}{(s - \rho)^r}.$$

The second numerator vanishes to order at least r , by the actual analytic factorization; its value after division at ρ is its r th derivative divided by $r!$. Starting with U_0 , these are explicit triangular recurrences. They are inverses on cluster jets arising from (4); arbitrary formal jets are not asserted to be admissible.

Conversely, agreement of moments through J gives agreement of Z to order t^{J+1} on the contour, also after s -differentiation. The denominators there stay nonzero, so the contour power sums, Newton coefficients, and monic polynomials agree to the same order. This proves the converse. The argument reproduces the portion of [2], H10–H13, needed here, and requires neither RH nor simplicity. $\square$

Theorem 4.2 (Finite spectral reconstruction). *The map from $\Delta(W_{m,A})$ to actual zero-cluster jets of order $K - 1$ in (4) is injective, with singleton fibres on its image. Its inverse is (6)–(7) followed by*

$$(8) \quad \ell_w(y) = \prod_{v \neq w} \frac{y - r_v}{r_w - r_v} = \sum_{j=0}^{K-1} \ell_{w,j} y^j, \quad \lambda_w = \sum_{j=0}^{K-1} \ell_{w,j} M_j.$$

For $K \geq 2$, order $K - 2$ does not suffice for all probability vectors. For $K = 1$, the unique weight is already known from $M_0 = 1$.

Proof. Distinct nodes were proved in Lemma 3.1. Thus every denominator in (8) is nonzero and $\ell_w(r_v)$ is the Kronecker delta. Integrating gives the inverse; the preceding lemma gives its moments.

For sharpness let $h_w = \prod_{v \neq w} (r_w - r_v)^{-1}$. Comparing the leading coefficient in Lagrange interpolation of y^j gives

$$\sum_w h_w r_w^j = 0 \quad (0 \leq j \leq K - 2), \quad \sum_w h_w r_w^{K-1} = 1.$$

The vectors $\lambda_w^\pm = 1/K \pm \varepsilon h_w$ are distinct strictly positive probabilities whenever $0 < \varepsilon < 1/(K \max_w |h_w|)$. They have equal moments through $K - 2$, hence identical cluster jets of that order, and distinct next moments. This proves the claimed lower bound. $\square$

This is an exact inverse, not a numerical conditioning bound. In particular, the shrinking guaranteed t -disc as A grows and the Lagrange coefficients remain present in the construction.

5. ARITHMETIC PREDICATES AND AN EXPLICIT THREE-WORD CALCULATION

For any function $f : W \rightarrow \mathbb{R}$, define its interpolating polynomial

$$H_f(y) = \sum_w f(w) \ell_w(y) = \sum_{j=0}^{K-1} h_j y^j.$$

Then the exact expectation is

$$(9) \quad \sum_w \lambda_w f(w) = \sum_{j=0}^{K-1} h_j M_j.$$

For odd moduli $q_1, \dots, q_b$, choose $f(w) = \prod_i \mathbf{1}_{B_w \equiv c_i \pmod{q_i}}$ to recover the joint residue mass. The product is evaluated on a single word, so no independence of the marginal tests is introduced. Pairwise coprimality is needed only to identify the tuple with one Chinese-remainder residue, not to define or recover this joint mass.

For $D > 0$, the choice $f(w) = \mathbf{1}_{D|B_w}$ recovers the mass of integral periodic candidates. With uniform weights, multiplying this mass by K gives the number of qualifying labelled words, including repetitions. It is not a count of primitive cycles. A particular predicate can require fewer moments if its interpolating polynomial has lower degree; the sharpness theorem concerns the entire probability vector.

For $(m, A) = (2, 4)$, $D = 7$ and the complete data are

w	B_w	r_w	$7 \mid B_w$
(1, 3)	5	19	no
(2, 2)	7	1	yes
(3, 1)	11	29	no

Thus

$$(10) \quad \lambda_{(2,2)} = \frac{M_2 - 48M_1 + 551}{504}.$$

Indeed, the relevant interpolant is $(y - 19)(y - 29)/((1 - 19)(1 - 29))$. Uniform weights give $M_1 = 49/3$, $M_2 = 401$, and mass $1/3$. The qualifying word repeats the known odd fixed point 1; the example is an exact transfer calculation.

To write (10) directly from a simple zero trajectory, let $\rho + c_1 t + c_2 t^2 + \dots$ be that trajectory. Then

$$M_1 = \frac{\zeta'(\rho)c_1}{\rho\zeta(\rho+1)},$$

$$M_2 = -\frac{2}{\rho(\rho+1)\zeta(\rho+2)} \left(\zeta'(\rho)c_2 + \frac{\zeta''(\rho)}{2}c_1^2 - M_1(\zeta(\rho+1) + \rho\zeta'(\rho+1))c_1 \right).$$

The general-multiplicity formulas (6)–(7) give the same mass without a simplicity assumption.

5.1. Maps of histories and the role of zero weights. For a map of finite labelled sets $\pi : W \rightarrow V$ (for example, deleting the last exponent), the induced probability map is

$$(\pi_*\lambda)_v = \sum_{w:\pi(w)=v} \lambda_w.$$

Its linear extension has kernel $\{h : \sum_{\pi(w)=v} h_w = 0 \text{ for all } v\}$; on probability vectors each nonempty fibre is exactly the set with these prescribed fibre sums. This computes the histories lost by taking prefixes. The two spectral encodings commute with this map after decoding: recover λ , apply these sums, and construct the target Hurwitz family using any specified distinct coordinates on V . This defines a map on the actual images, not on arbitrary analytic germs.

To transport values without decoding each component separately, interpolate $w \mapsto f(\pi(w))$ in (9). This preserves precisely the same expectations. If a set $S \subseteq W$ also records histories retained despite zero weight, the input is (S, λ) with $\text{supp } \lambda \subseteq S$. The spectral family depends only on λ . Its fibre in this enlarged input consists of the possible $S \supseteq \text{supp } \lambda$. Retaining S separately is therefore necessary to distinguish supported zero weight from absence, exactly as in the Split Zero construction. It is not information present in the measure alone.

6. MATHEMATICAL USE AND VERIFICATION

The construction supplies faithful finite spectral coordinates for the same joint arithmetic tests used in the Collatz residue workbench. It also gives explicit witnesses when a chosen projection loses history. The Hurwitz family is built from the history weights themselves; recovering them from its zeros does not discover an unknown distribution of infinite positive-integer trajectories. A quantitative use for descent would need additional estimates about those trajectories. No such estimate is proved here, and no equivalence between RH and Collatz is inferred from using a zeta deformation.

The accompanying standard-library certificate checks finite parity-word compositions, exact residue realizations, polynomial reconstruction, the specialization collision, Lagrange inverses, and the displayed packet calculation using integer and rational arithmetic. The general assertions are proved above. The analytic zero-cluster argument is not Lean-certified or established by sampling numerical zeta zeros. The source records and independent review identify which part of the preceding Split Zero programme was used.

REFERENCES

- [1] *Support-Idempotent Homogenization for a Split-Zero Collatz F-Series Frame*, local technical note, April 2026, Sections 1, 4, 5, and 7. The exact source file and SHA256 are in the accompanying source manifest. The present note uses the displayed finite branch matrices and prefix formula; it does not rely on all claims in that note.
- [2] Split Zero research programme, *Historical Hurwitz reconstruction and the complete Laurent jet bridge*, equations H1–H13, especially H10–H13. Public source snapshot, 13 September 2026: `zeta-function-research-reader`, commit `16fc4dbb817b`.

- [3] Kokuno Yumeto, *Weighted Odd-Step Paths, Survivor Coding, Arithmetic Groupoids, and Toric Period Morphisms: A Research Companion for the Collatz Map*, Sections 2 and 4, working edition, 4 September 2026. Collatz workbench.
- [4] NIST Digital Library of Mathematical Functions, equation 25.11.10, Hurwitz zeta Taylor expansion. Accessed 13 September 2026.